SCTR's Pune Institute of Computer Technology
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PBL Presentation

DETECTION OF MILK ADULTERATION USING IOT AND ML MODEL

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Introduction

- Milk is consumed daily by millions
  Staple food for all age groups
- Quality directly impacts human health
  Poor milk can cause illness and infections
- Milk adulteration is increasing rapidly
  Demand and profit motives drive malpractice
- Water and harmful chemicals are added
  Urea detergents starch and formalin are common
- Causes health risks and nutritional loss
  Reduces protein value and may be toxic
- Laboratory testing is costly and slow
  Requires expensive equipment and skilled staff

Literature Survey

Aqeel et al. (2025) used hyperspectral imaging and ML for milk adulteration classification with high accuracy, but relies on expensive imaging and lab setups.

- Liang et al. (2025) applied portable NIR spectroscopy with chemometrics to detect water adulteration with excellent discrimination, yet still uses specialized NIR equipment.
- Zhang et al. (2025) developed a holographic sensor for rapid milk adulteration detection through optical grating shifts, offering speed but requiring novel optical materials.
- Mamgain (2024) demonstrated machine learning classification of milk adulterants from evaporative image patterns using CNNs, highlighting the potential of AI-based detection.
- Durgun (2024) reported a portable multispectral sensor system with logistic regression for real-time detection of starch adulteration, showing low-cost real-time potential but limited spectrum range.

MOTIVATION

Milk is consumed every day

Used by children adults and elderly

- Adulteration affects public health

  Can cause serious diseases and toxicity

- Manual detection is unreliable

  Impurities cannot be seen by eye

- Laboratory tests take too much time

  Results are delayed and inconvenient

- Testing equipment is very expensive

  Small farmers cannot afford it

- Rural areas lack quality testing facilities

  Unsafe milk often reaches consumers

- Need simple portable smart solution

Quick on site detection is essential

PROBLEM DEFINITION AND OBJECTIVES

Problem Definition

- Milk adulteration is widely increasing
- Harmful chemicals reduce milk quality
- Manual checking is not reliable
- Lab testing is slow and expensive
- No portable on site detection device

Objectives

- Design a portable testing system
- Integrate spectral pH conductivity sensors
- Develop compact PCB with ESP32
- Collect dataset from real samples
- Train edge AI model for detection
- Provide instant and accurate results

PROPOSE METHOD AND TECHNIQUES

Multi sensor based detection approach

- Spectral sensor (AS7343 14 CHANNEL SPECTRAL SENSOR) measures optical properties
- pH sensor monitors acidity changes
- Conductivity sensor measures ionic content
- ESP32 collects and processes data
- Features extracted from sensor readings
- Edge MLmodel classifies milk quality
- Output displayed as pure or adulterated

SOCIETY CONTRIBUTIONS

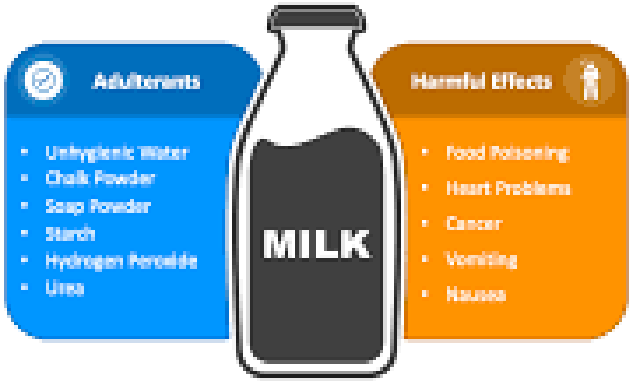
Ensures safe and pure milk for consumers

- Protects public health and food safety
- Reduces risks from harmful adulterants
- Enables on site milk quality testing
- Helps farmers and dairy vendors instantly
- Eliminates dependence on laboratories
- Saves time cost and effort
- Promotes affordable testing technology
- Supports rural and small scale dairies

RESULT ANALYSIS

Pure milk samples tested as baseline

- Adulterated samples prepared at different concentrations
- Spectral pH and conductivity values recorded
- Clear variation observed between pure and adulterated milk
- Dataset created from real sensor readings
- AI model trained using collected data
- High detection accuracy achieved
- Fast real time prediction obtained
- Stable and consistent performance observed



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